

Antiviral Agents

Introduction

- ❖ Viruses are the smallest infectious agents
- ❖ They vary greatly in structure, genome organization and expression, and strategies of replication and transmission
- ❖ Viruses are obligate intracellular parasites;
 - ✓ their replication depends primarily on synthetic processes of the host cell.
 - to be effective, antiviral agents must either block viral entry into or exit from the cell or be active inside the host cell.
 - nonselective inhibitors of virus replication may interfere with host cell function and result in toxicity

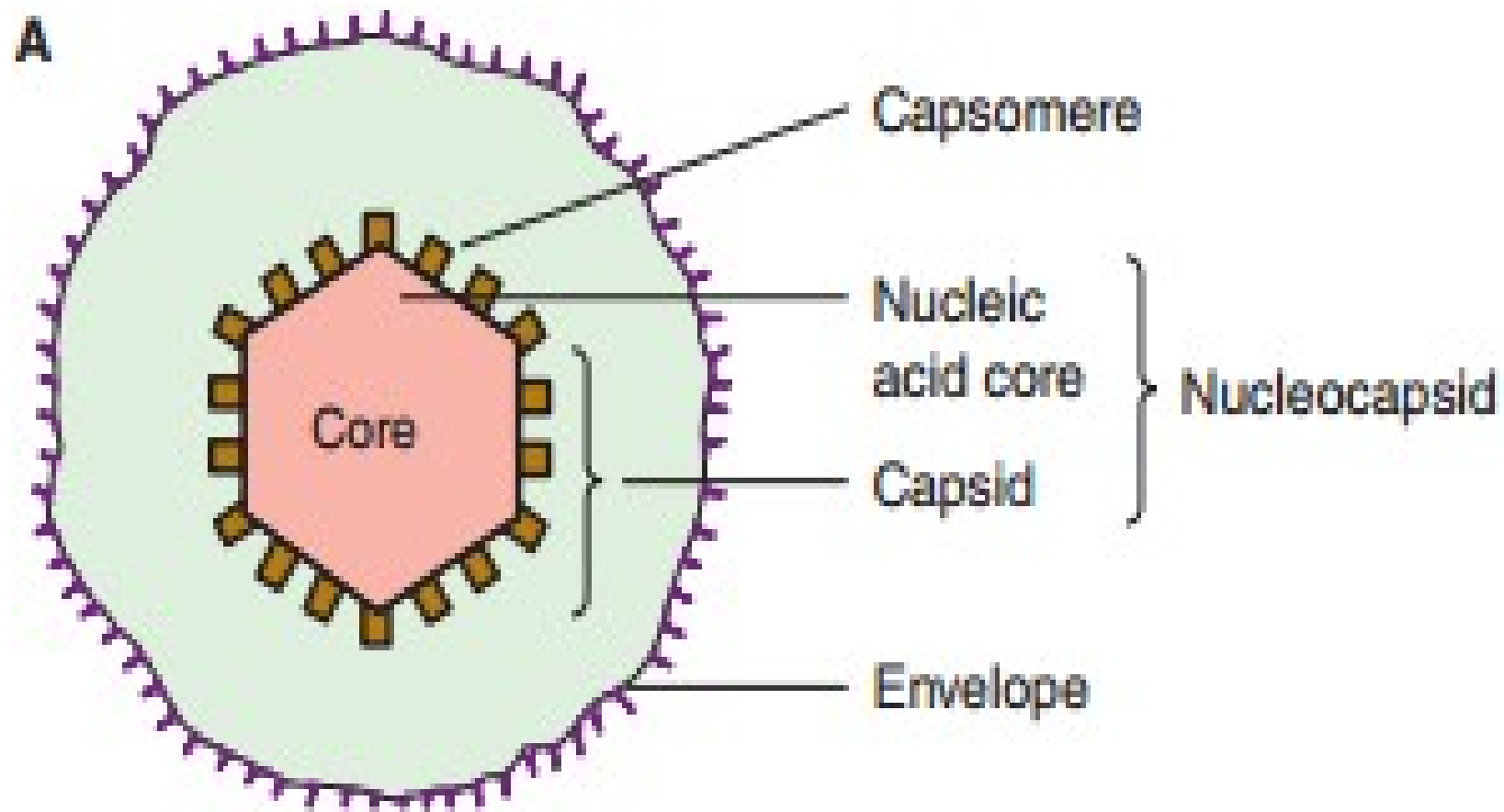


Figure: viral components

Introduction...

❖ Viruses can be

DNA virus

✓ **Example:** poxviruses (smallpox), herpesviruses (chickenpox, shingles, oral and genital herpes); adenoviruses (conjunctivitis, sore throat); hepadnaviruses (HBV); and papillomaviruses (warts)

❖ Most DNA viruses enter the host cell nucleus

- ✓ viral DNA is transcribed into mRNA by host cell polymerase;
- ✓ mRNA is translated in the usual host cell fashion into virus-specific proteins.

RNA virus

- Picornaviruses, Hepeviruses, Retroviruses, Coronaviruses

Introduction...

- ❖ Progress in antiviral chemotherapy began in the early 1950s,
- ❖ the search for anticancer drugs generated several new compounds capable of inhibiting viral DNA synthesis.
 - ✓ 5-iododeoxyuridine and trifluorothymidine,
 - had poor specificity.....toxic for systemic use but used topically for the treatment of **herpes keratitis**
- ❖ Antiviral therapy is now available for
 - ✓ herpes simplex virus (HSV),
 - ✓ cytomegalovirus (CMV),
 - ✓ varicella zoster virus (VZV),
 - ✓ hepatitis C virus (HCV), hepatitis B virus (HBV),
 - ✓ influenza,
 - ✓ human immunodeficiency virus (HIV),
 - ✓ and respiratory syncytial virus (RSV).

Introduction...

Common property of antiviral agents

- ✓ they are active only against replicating viruses and do not affect latent virus
- ✓ Virus replication cycle and targets

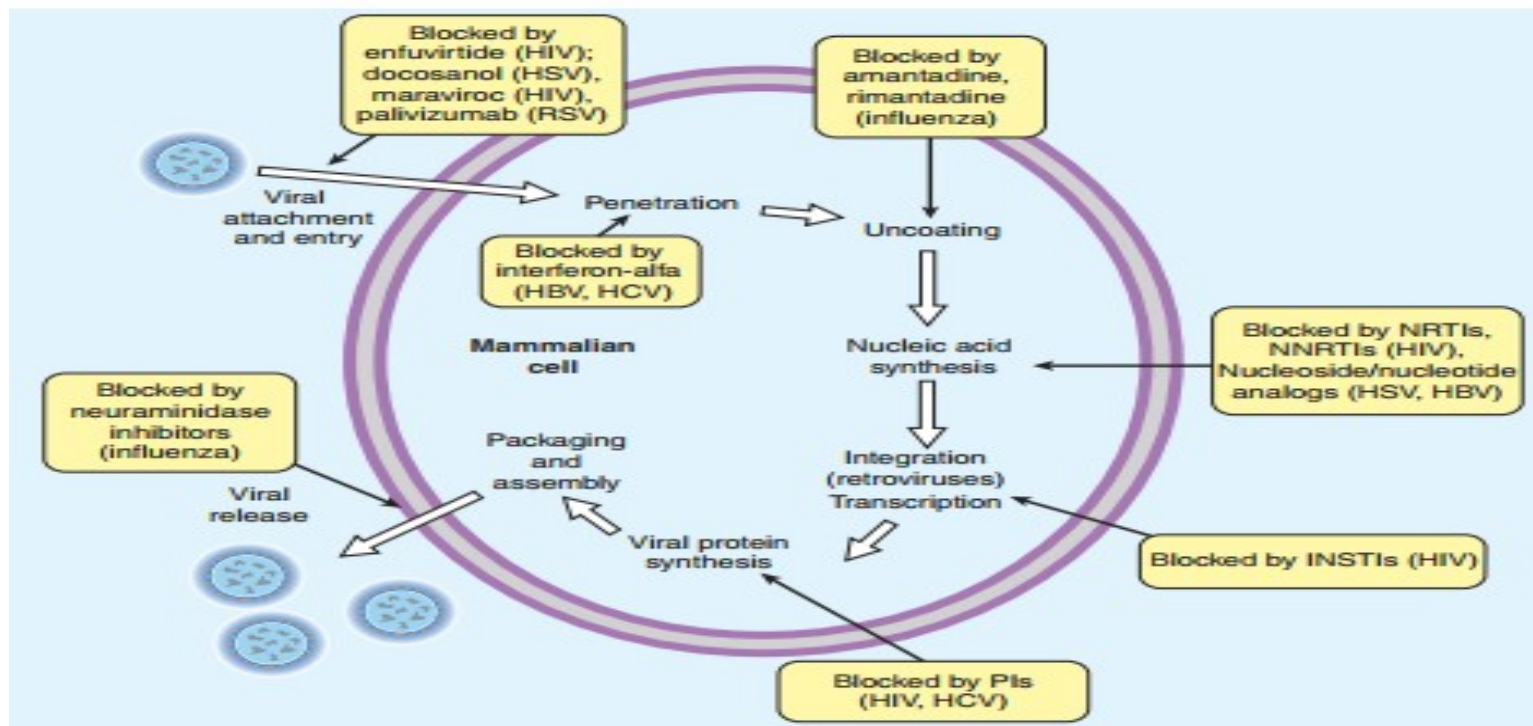


Figure: The major sites of antiviral drug action

Agents to treat herpes simplex virus (HSV) & varicella-zoster virus (VZV) infections

Drugs: Acyclovir, valacyclovir, and famciclovir.

- They have similar mechanisms of action and
- comparable indications for clinical use
- ❖ The efficacy of the antiherpes agents in **orolabial** herpes is generally less than that in **anogenital herpes**

Agents to treat herpes simplex virus (HSV) & varicella-zoster virus (VZV) infections...

Table : Agents to treat or prevent herpes simplex HSV and VZV infections.

Agent	Treatment of First Episode	Treatment of Recurrent Episodes	Suppression
Genital Herpes			
Acyclovir, oral ¹	400 mg tid × 7–10 days or 200 mg 5 times daily	800 mg tid × 2 days or 800 mg bid × 5 days or 400 mg tid × 5 days	400–800 mg bid-tid ²
Famciclovir, oral ¹	250 mg tid × 7–10 days	1000 mg bid × 1 day or 125 mg bid × 5 days or 500 mg once then 250 mg bid × 2 days ²	250–500 mg bid ²
Valacyclovir, oral ¹	1000 mg bid × 10 days	500 mg bid × 3 days or 1 g qd × 5 days	500–1000 mg qd-bid ²
Orolabial herpes			
Acyclovir, oral ¹	400 mg tid × 7–10 days or 200 mg 5 times daily	200–400 mg 5 times daily × 5 days	400–800 mg bid-tid ²
Famciclovir, oral ¹	500 mg tid × 7–10 days	1500 mg once or 750 mg bid	500 mg bid
Valacyclovir, oral ¹	1 g bid × 7–10 days	2 g bid × 1 day	500–1000 mg qd
Acyclovir, topical (5% cream)		5 times daily × 4 days	
Docosanol, topical (10% cream)		5 times daily	
Penciclovir, topical (1% cream)		Every 2 h while awake	
Herpes proctitis. treatment			

Acyclovir..

- ❖ It is an acyclic **guanosine derivative**
- ❖ activity against HSV-1, HSV-2, and VZV
- ❖ Comparative potency
 - ✓ **10* more potent against HSV-1 & HSV-2** than VZV
- ❖ Acyclovir requires three phosphorylation steps for activation.
- ❖ First converted first to the **monophosphate derivative (by virus specified thymidine kinase)** then to the di- and triphosphate compounds by host cell enzymes

Acyclovir

- ❖ Acyclovir triphosphate inhibits viral DNA synthesis by two mechanisms:
 - ✓ competition with **deoxyGTP** for the viral DNA polymerase,
 - resulting in binding to the **DNA template as an irreversible complex**; and chain **termination** following
- ❖ **incorporation into the viral DNA**

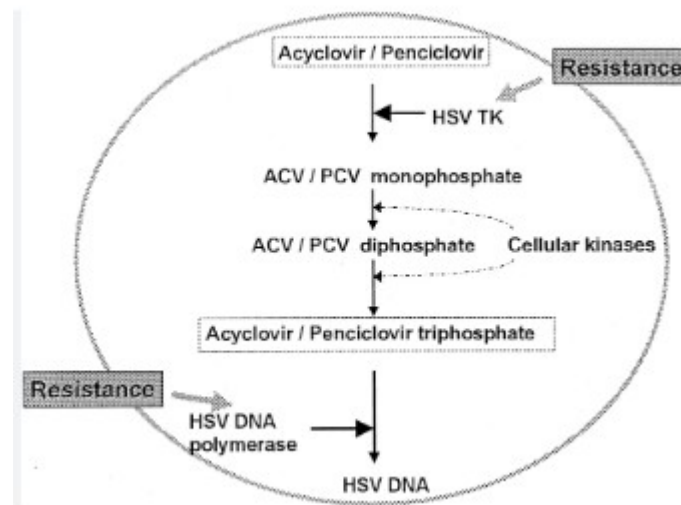


Figure: Acyclovir and Penciclovir

Pharmacokinetics and uses

- ❖ bioavailability is low (15–20%) and is unaffected by food.
- ❖ **Topical formulations** produce high concentrations in **herpetic lesions**,
 - ✓ but systemic concentrations are **undetectable by this route**
- ❖ Clearance.... primarily by glomerular filtration and tubular secretion.
- ❖ The half-life
 - ✓ Normal individual....2.5-3hr
 - ✓ in patients with anuria..... 20 hours

Pharmacokinetics and uses..

- ❖ Acyclovir is available in IV form and used
 - ✓ herpes simplex encephalitis,
 - ✓ neonatal HSV infection, and
 - ✓ serious HSV or VZV infections
 - ✓ In immunocompromised patient
 - intravenous acyclovir reduces the incidence of cutaneous and visceral dissemination
- ❖ In neonates with central nervous system HSV,
 - ✓ **oral acyclovir** suppression for **6 months** following acute treatment improves neurodevelopmental outcomes.

Pharmacokinetics and uses

❖ **Topical acyclovir cream**

- ✓ **less effective than oral therapy for primary HSV infection.**
- ✓ **no benefit in treating recurrent genital herpes**

❖ **Resistance** to acyclovir can develop in HSV or VZV through

- ✓ alteration in either the viral thymidine kinase or
- ✓ the DNA polymerase
- ✓ **Cross resistance** can happen b/n acyclovir and valacyclovir, famciclovir, and ganciclovir.
- ✓ But not b/n acyclovir and foscarnet, cidofovir, and trifluridine
 - **do not require activation by viral thymidine kinase** and therefore have
 - **preserved activity against** the most prevalent acyclovir-resistant strains

Side effects

- ❖ Acyclovir is generally well tolerated,
- ❖ nausea, diarrhea, and headache may occur.
- ❖ IV formulation is associated with
 - ✓ reversible renal toxicity (ie, crystalline nephropathy or
 - ✓ interstitial nephritis) or
 - ✓ neurologic effects (eg, tremors, delirium, seizures).

Valacyclovir

- ❖ Valacyclovir is the l-valyl ester of acyclovir
 - rapidly converted to acyclovir after oral administration via first-pass enzymatic hydrolysis in **the liver and, intestine**

Valacyclovir

- ❖ After conversion...it results 3-5 serum con of that of acyclovir
- ❖ Oral bioavailability is 54–70%,
- ❖ Elimination half-life is 2.5–3.3 hours.

Side effect

- ❖ Valacyclovir is generally well tolerated, although nausea, headache, vomiting, or rash may occur.
- ❖ At high doses.....confusion, hallucinations, and seizures have been reported.AIDS
- ❖ GI intolerance, thrombotic thrombocytopenic purpura/hemolytic-uremic syndromein AIDS patients who received high-dosage valacyclovir chronically

Famciclovir

- ❖ It is the diacetyl ester prodrug of **6-deoxypenciclovir**, an acyclic **guanosine analog**.
- ❖ After oral administration,
- ❖ It rapidly **deacetylated** and oxidized by first-pass metabolism to **penciclovir**.
- ❖ Like acyclovir, **activation by thymidine kinase is required**
- ✓ **Produce competitive inhibition** of the viral DNA polymerase to block DNA synthesis.
- ✓ But unlike acyclovir, **penciclovir does not cause chain termination**.

Famciclovir...

- ❖ Penciclovir triphosphate has lower affinity for the viral DNA polymerase than acyclovir triphosphate, **but it achieves higher intracellular concentrations.**
- ❖ Oral bioavailability... 70%.
- ❖ excreted primarily in the urine
- ❖ Half life is prolonged, at 7–20 hours

S/E

- ❖ Oral famciclovir is generally well tolerated, although headache, nausea, or diarrhea may occur.

Penciclovir

- ❖ The **guanosine** analog penciclovir, the active metabolite of famciclovir,
- ❖ available for topical use, cream (1 %).

Adverse effects are uncommon, other than application site reactions in ~1%

Docosanol

- ❖ It is a saturated 22-carbon aliphatic alcohol
- ❖ It inhibits fusion between the **host cell plasma membrane** and the **HSV** envelope,
 - prevent viral entry into cells and subsequent viral replication.
- ✓ Topical (10% cream) application can shorten the healing time

Trifluridine

- ❖ Trifluridine (trifluorothymidine) is a fluorinated pyrimidine nucleoside that inhibits viral DNA synthesis in
 - HSV-1,
 - HSV-2,
 - CMV and
 - some adenoviruses.
 - ❖ MOA.....after it is is phosphorylated intracellularly by host cell enzymes,
 - Then competes with **thymidine triphosphate** for incorporation by the viral DNA polymerase
- Drawback**...it also incorporates in **host DNA** (prevents its systemic use).

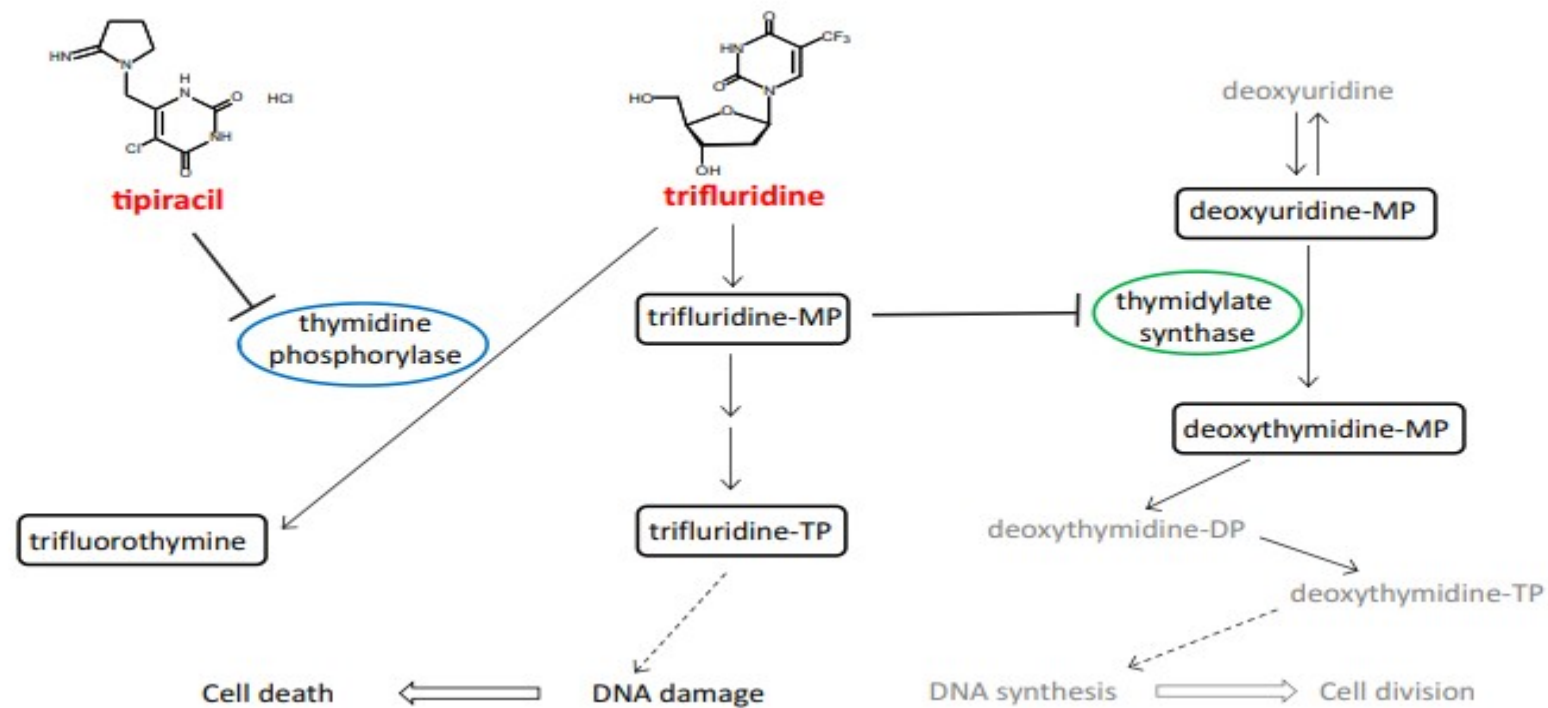


Figure 1: Mechanism of trifluridine/tipiracil-mediated cytotoxicity

Trifluridine...

Uses

- ❖ Application of a 1% solution is effective in
- ❖ **keratoconjunctivitis** and recurrent **epithelial keratitis** due to HSV-1 or HSV-2.
- ❖ In combination with **interferon-alfa** or alone
- ✓ has been used successfully in the treatment of acyclovir-resistant **HSV infections**.

Agents used to treat cytomegalovirus

❖ **CMV infections** occur primarily in the setting of **advanced immunosuppression** and are typically due to **reactivation of latent infection**.

❖ Dissemination of infection results in

—end-organ disease, including **retinitis, colitis, esophagitis, CNS disease, and pneumonitis**.

Drugs in this class: Ganciclovir, Valganciclovir, Foscarnet and Cidofovir

Agent	Routes of administration	Uses	Recommended adult dose
Valganciclovir	Oral	CMV retinitis treatment	Induction: 900 mg bid × 21 da
			Maintenance: 900 mg daily
	Ora	CMV prophylaxis (transplant patients)	900 mg daily
Letermovir	Oral, intravenous	CMV prophylaxis (transplant patients)	480 mg once daily orally or IV over 1 hour
Ganciclovir ¹	Intravenous	CMV retinitis treatment	Induction: 5 mg/kg q12h × 14–21 days
			Maintenance: 5 mg/kg/d or 6 mg/kg five times per week
Foscarnet	Intravenous	CMV retinitis treatment	Induction: 60 mg/kg q8h or 90 mg/kg q12h × 14–21 day
			Maintenance: 90–120 mg/kg/d
Cidofovir	Intravenous	CMV retinitis treatment	Induction: 5 mg/kg/wk × 2 weeks
			Maintenance: 5 mg/kg every

Ganciclovir

- ❖ Ganciclovir is an acyclic **guanosine analog** that requires activation by triphosphorylation
- ❖ Initial phosphorylation
 - catalyzed by the virus-specified protein kinase phosphotransferase **UL97** in **CMV-infected cells**.
 - The activated compound
 - competitively inhibits viral DNA polymerase and causes**
 - **termination of viral DNA elongation**
- ❖ Its activity against CMV is up to **100 times** greater than that of **acyclovir**.
- ❖ Administered intravenously.
 - The bioavailability of **oral ganciclovir** is poor and it is **no longer available in the USA**.

Ganciclovir....

- ❖ Ganciclovir gel is used..for the treatment of **acute herpetic keratitis..**

PK

- half-life.... 4 hours,
- Ganciclovir is readily cleared by hemodialysis

Indications

- ❖ **IV ganciclovir** has been shown to delay progression of **CMV retinitis** in immunocompromised patients.
- ❖ Dual therapy with **foscarnet** and **ganciclovir** is more effective in **delaying progression of retinitis** than either drug alone in patients with AIDS
 - But ADR is compounded
- ❖ **IV ganciclovir is also used to treat**
 - CMV colitis, esophagitis, and in combination with IV cytomegalovirus immunoglobulin.. for **pneumonitis**
- ❖ **IV ganciclovir followed by either oral ganciclovir or high-dose oral acyclovir**
 - reduces the risk of CMV infection in transplant recipients.

❖ **IV Ganciclovir**

- Reduce hearing loss in symptomatic congenital neurologic CMV disease

❖ **Ganciclovir due to its effect on HHV-8.**

- The risk of Kaposi sarcoma is reduced in AIDS patients receiving (long-term treatment)

Mechanism of resistance

- ❖ Resistance to ganciclovir increases with duration of use
- ❖ mutation, in UL97, results in decreased levels of the triphosphorylated (ie, active) form of ganciclovir...**more common**
- ❖ UL54 mutation in DNA polymerase results in **higher levels of resistance** and **potential cross-resistance** with cidofovir and foscarnet....**Less common**

Adverse Effects and Drug interaction

- ❖ **Myelosuppression....Most common but irreversible**
- ❖ Other **potential adverse effects...** nausea, diarrhea, fever, rash, headache, insomnia, and peripheral neuropathy
- ❖ Central nervous system toxicity (confusion, seizures, psychiatric disturbance) and hepatotoxicity

Adverse Effects and Drug interaction

❖ Intravitreal ganciclovir

–vitreous hemorrhage and retinal detachment.

❖ **probenecid + ganciclovir**

–rise in patients concurrently taking probenecid

❖ Concurrent use of ganciclovir with **didanosine**

–increased levels of didanosine.

Valganciclovir

❖ Valganciclovir is an **L-valyl ester** prodrug of **ganciclovir**

❖ It exists as a mixture of two diastereomers.

–After oral administration, both diastereomers are rapidly hydrolyzed to ganciclovir by esterases in the intestinal wall and liver.

- ❖ Valganciclovir has a bioavailability of **60%** and should be taken with food.
- ❖ The major route of elimination is **renal, through GFR** and active **tubular secretion**.
- ❖ Plasma concentrations of valganciclovir are reduced ~**50%** by hemodialysis.
- ❖ Valganciclovir **is as effective as IV** ganciclovir for the treatment of CMV retinitis
- ❖ indicated for the prevention of CMV disease **in high-risk solid organ and bone marrow transplant recipients**.

Adverse effects and Drug interaction

- ❖ drug interactions, and resistance patterns are the same as with ganciclovir.

Foscarnet

- ❖ Foscarnet (phosphonoformic acid) is an inorganic pyrophosphate analog
- ❖ It inhibits
 - herpesvirus DNA polymerase & RNA polymerase,
 - HIV reverse transcriptase directly **without requiring activation by phosphorylation.**
- ❖ Foscarnet blocks the pyrophosphate binding site of **d/f enzymes** and inhibits **cleavage of pyrophosphate from deoxynucleotide triphosphates.**
- ❖ It is available in an IV formulation only;
 - poor oral bioavailability and
 - gastrointestinal intolerance preclude oral use.

Foscarnet

- ❖ plasma half-life is 3–7 hours,
 - up to 30% of foscarnet may be deposited in bone, with a half-life of **several months**
 - Its clinical repercussion is not known
- ❖ Serum drug concentrations are reduced ~50% by hemodialysis.

Indications

- ❖ end-organ CMV disease (ie, retinitis, colitis, and esophagitis), **including ganciclovir-resistant disease;**
- ❖ **acyclovir-resistant HSV and VZV infections.**

Foscarnet

- ❖ The main toxicity is renal impairment
 - Saline preloading helps prevent nephrotoxicity
 - The dosage must be titrated according to creatinine **clearance** before **each infusion**.
 - Use of an infusion pump to **control the rate** of infusion is important to **prevent toxicity**,
- ❖ **large volumes of fluid are required because of the drug's poor solubility.**
- ❖ In combination with ganciclovir compare to each alone
 - delay progression of retinitis; but
 - toxicity is also increased
- ❖ As with ganciclovir, it decreases in the incidence of Kaposi sarcoma

Mechanism resistance

- ❖ Resistance to foscarnet in HSV and CMV is due to **point mutations** in the DNA polymerase gene and
 - associated with prolonged or repeated exposure to the drug.
 - Although foscarnet-resistant CMV isolates are typically cross-resistant to ganciclovir, foscarnet is active against **ganciclovir-** and **cidofovir-resistant isolates of CMV**.
 - Mutations in the HIV-1 reverse transcriptase gene

Adverse drug Reaction

- ❖renal impairment, hypo- or hypercalcemia, hypo- or hyperphosphatemia, hypokalemia, and hypomagnesemia.

Adverse drug Reaction

- ❖ The risk of severe hypocalcemia is increased with concomitant use of pentamidine,
 - Due to chelation of divalent cations,.
- ❖ Genital ulcerations
 - due to high levels of ionized drug in the urine.
- ❖ Nausea, vomiting, ↑ anemia (especially during concurrent use of zidovudine), liver enzymes, and fatigue
- ❖ Central nervous system (headache, hallucinations, and seizures);
 - seizures** may be increased with concurrent use of **imipenem**

Cidofovir

- ❖ Cidofovir is an acyclic cytosine nucleotide
- ❖ In contrast to **ganciclovir**, phosphorylation of cidofovir to the active diphosphate is **independent of viral enzymes**
 - activity is maintained against thymidine kinase-deficient or -altered strains of CMV or HSV.
- ❖ Cidofovir diphosphate acts both as a **potent inhibitor** and as an **alternative substrate** for viral DNA polymerase.....Competes with **dCTP**
 - competitively inhibiting DNA synthesis and becoming incorporated into the viral DNA chain.
 - Cidofovir-resistant is cross-resistant with **ganciclovir** but retain **susceptibility to foscarnet**.

Cidofovir.....mechanisms of drug resistance

❖ Cidofovir resistance in CMV is due to mutations in **viral DNA polymerase**

- **Highly ganciclovir-resistant** CMV isolates that possess DNA **polymerase and UL97 kinase mutations** are resistant to cidofovir
- Some foscarnet-resistant CMV isolates show cross-resistance to cidofovir, and
 - triple-drug-resistant variants with DNA polymerase mutations occurs

Cidofovir

- ❖ Parent drug half life2.6 hours,
- ❖ Active metabolite (**cidofovir diphosphate**) has a prolonged half-life of **17–65 hours**,
 - Do not require frequent dosing
- ❖ Cerebrospinal fluid penetration is poor.
- ❖ Elimination is by active **renal tubular secretion**.
 - High-flux hemodialysis reduces serum levels of cidofovir by approximately 75%.

Cidofovir...

- ❖ IV cidofovir is effective for the treatment of CMV retinitis
- ❖ cidofovir is administered with high-dose probenecid (2 g at 3 hours before the infusion and 1 g at 2 and 8 hours after),
 - Block active tubular secretion and
 - decreases nephrotoxicity.
- ❖ **Dose** is **given** according to **CrCl** or aggressive adjunctive hydration is required.
 - Initiation of cidofovir therapy is C/I in patients with existing renal insufficiency.
- ❖ Direct **intravitreal** administration of cidofovir is not recommended due to ocular toxicity.

Cidofovir...

Adverse Effect

- ❖ The primary adverse effect of intravenous cidofovir is a dose-dependent proximal tubular nephrotoxicity,
 - can be reduced with normal saline prehydration
 - Proteinuria, azotemia, metabolic acidosis, and Fanconi syndrome may occur
 - Concurrent administration of other potentially nephrotoxic agents should be avoided.
- ❖ Other potential adverse effects include uveitis, ocular hypotony, and neutropenia (15–24%).

Letermovir

- ❖ Letermovir is indicated for CMV prophylaxis in adult CMV-seropositive recipients of an allogenic hematopoietic stem cell transplant.
- ❖ available in both **oral** and **IV** formulations.
- ❖ It has a novel mechanism of action,
 - inhibiting the CMV DNA terminase complex required for **DNA packaging** and processing and **thus replication**.
- ❖ Its antiviral activity is highly specific to CMV and is currently the most active molecule against CMV,
 - with preserved activity against CMV isolates that are resistant to other antiviral agents.

Adverse effects and drug interaction

- ❖ Potential adverse effects include
 - tachycardia, atrial fibrillation, nausea, and diarrhea.
- ❖ it is a moderate CYP3A4 and OATP1B1/3 inhibitor and
- ❖ a substrate of OATP1B1/3,
 - there are multiple potential drug-drug interactions.
- ❖ pimozone and ergot alkaloids is contraindicated.
- ❖ The dose of letermovir should be decreased if administered concomitantly with cyclosporine.

Introduction

- ❖ Influenza virus strains are classified by
 - their core proteins (ie, A, B, or C),
 - species of origin (eg, avian, swine), and
 - geographic site of isolation.
- ❖ **Influenza A,**
 - the only strain that causes pandemics,
 - influenza A viruses can infect a variety of animal hosts including birds in addition to human
 - classified into 16 H (hemagglutinin) and 9 N (neuraminidase) subtypes based on surface proteins. Although
 - Specieses that are circulating among worldwide populations include H1N1, H1N2, and H3N2
- ❖ Influenza B
 - Infect only human
- ❖ Influenza C virus infections
 - cause mild illness and are not thought to cause human epidemics

❖ Species of origin

Avian

- Avian influenza refers to the disease caused by infection with avian (bird) influenza (flu) ...it is **type A viruses**
- occur naturally among wild **aquatic birds** worldwide and can infect **domestic poultry** and other bird and animal species
- do not normally infect humans. However, sporadic human infections with avian flu viruses have occurred

❖Swine flu

- it is a virus that pigs can get.
- People do not usually get swine flu
- In 2009 a strain of swine flu called H1N1 infected many people around the world.

Virus life cycle

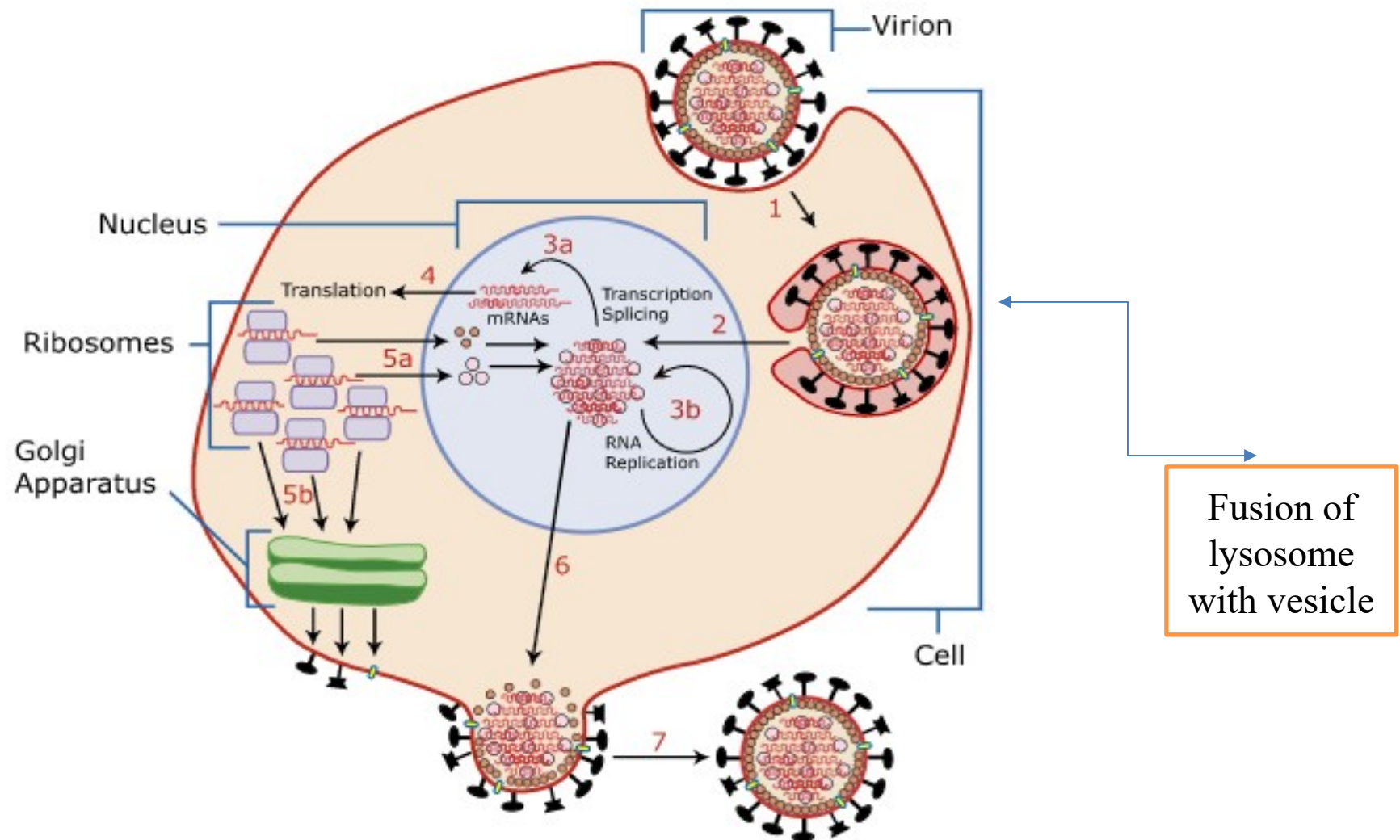


Figure: Life cycle of Influenza Virus

Introduction

- ❖ Until 2018, only five anti-influenza drugs were
 - three **neuraminidase** inhibitors (oral oseltamivir, inhaled zanamivir, IV peramivir) and
 - two **adamantanes** (amantadine, rimantadine).
- ❖ The recent availability of **baloxavir**, a selective inhibitor of **influenza cap-dependent endonuclease**,
 - offers an additional option for the treatment of influenza.
- ❖ **Adamantanes** are becoming almost **less useful**
 - The advent of the neuraminidase inhibitors with activity against both influenza A and B (adamantine active only against **influenza A** only by the), and
 - low level of **resistance** (vs >99% resistance to the adamantanes among both influenza H3N2 and influenza H1N1),

Although rates of resistance to neuraminidase inhibitors is low

- the emergence of more widespread neuraminidase inhibitor resistance remains a threat but
- Baloxavir **retains in vitro activity** against **neuraminidase** inhibitor-resistant viruses, as well as highly pathogenic **avian influenza** viruses.

Baloxavir marboxil

- ❖ It is a prodrug & converted by hydrolysis to the active baloxavir,
- ❖ Baloxavir is a cap-dependent endonuclease inhibitor
- ❖ MOA... interferes with **viral RNA transcription** and **blocks virus replication**.
 - Thus it acts earlier in the viral replication cycle than does oseltamivir.
- ✓ it binds to a different part of the virus than the neuraminidase inhibitors,
 - Synergism when it is combined with these agents

Baloxavir marboxil...

- ❖ The drug has activity against both **influenza A** and **influenza B**
- ❖ Due to its **long half-life (80 hours,)**given orally as a **single-dose** treatment.
- ❖ Treatment should be initiated within **48 hours** of the onset of symptoms
 - dose of **40 mg** or **80 mg**, depending on body weight.
- ❖It can decreased duration of symptoms
- ❖Compared to **oseltamivir**, **Baloxavir** produces
 - similar time to alleviation of symptoms
 - superior reduction in viral load at 1 day by baloxavir and
 - It is More is tolerable
- ❖**Baloxavir** remains effective against oseltamivir-resistant viruses and **vice versa**,

Baloxavir marboxil...

PK, drug interaction and effect of food

- ❖ It can be taken **with or without food** and
- ❖ Protein binding 93% bound to plasma proteins.
- ❖ Metabolism..... metabolized via **UGT1A3** with a minor contribution from CYP3A4.

Drug and food interaction

- ❖ **polyvalent cation-containing** products (eg, laxatives, antacids, calcium supplements) are contraindicated
 - due to a resultant decrease in baloxavir plasma concentrations.
- It should not be taken with dairy products.

Oseltamivir & zanamivir

- ❖ They are neuraminidase and analogs of sialic acid,
- ❖ interfere with release of progeny influenza A and B virus from infected host cells,
 - Halt the spread of infection within the respiratory tract.

MOA

- ❖ **competitively and reversibly interact** with the active **enzyme site** to inhibit viral **neuraminidase activity**
 - resulting in **clumping** of newly released influenza virions to each other and to the **membrane** of the infected cell.
- ❖ Early administration is crucial because
 - replication of influenza virus peaks at **24–72 hours** after the onset of illness..

Oseltamivir...

❖ It is oral pro-drug

–activated by hepatic esterases and widely distributed throughout the body.

❖ PK

–Bioavailability..... ~80%,

–plasma protein binding is low,

–The half-life of oseltamivir is 6–10 hours

–, and excretion is by glomerular filtration and tubular secretion.

–**Food doesn't affect the absorption**

Oseltamivir...

Dose

- ❖ Initiation of a 5-day course of therapy within 48 hours after the onset of illness
(**75 mg twice daily**) modestly
 - decreases the duration of **symptoms**, as well as
 - duration of **viral shedding**
- ❖ Once-daily prophylaxis (75 mg once daily) is 70–90% effective in **preventing disease after exposure.**
- ❖ **In declining renal function, oseltamivir carboxylate** (active metabolite)
concentration increases
 - **should be adjusted in patients with renal insufficiency**

Oseltamivir...

Adverse effects

- ❖ Potential adverse effects include nausea, vomiting, and headache.
- ❖ Other ... Fatigue and diarrhea
 - more common with prophylactic use.
- ❖ Rash (rare).
- ❖ Neuropsychiatric events (self-injury or delirium) in adolescent

Zanamivir

❖ Zanamivir to the respiratory tract via inhalation

- **10-20 % reaches the site**
 - the remainder is deposited in the oropharynx
 - The concentration of the drug in the respiratory = 1000*50 % inhibitory concentration for neuraminidase
- ❖ pulmonary half-life is **2.8** hours.

Zanamivir...

❖ Potential adverse effects include

- cough,
- bronchospasm (occasionally severe),
- reversible decrease in pulmonary function, and
- transient **nasal** and **throat** discomfort.

•Because of the above ADE **Zanamivir** is not recommended for patients with underlying airway disease

Peramivir

- ❖ Peramivir is a cyclopentane analog,
- ❖ It has activity against both influenza **A** and **B** viruses, and is
- ❖ Single dose... **600-mg IV dose**
 - Approved for the treatment of acute **uncomplicated influenza in adults.**
 - early treatment is promoted (ie, within 48 hours).

PK

- ❖ Less than 30% of peramivir is protein-bound.
- ❖ **Not metabolized** and the major **route of elimination is the kidney.**
 - Dose adjustment is required for renal insufficiency.
- ❖ Half-life...after IV administration is **~20 hours.**

Peramivir

Adverse Effects

- ❖ Diarrhea (main),
- ❖ serious skin reactions (eg, Stevens-Johnson syndrome, erythema multiforme)....Rare
- ❖ Psychiatric an increased risk of hallucinations, delirium

Amantadine & Rimantadine

- ❖ Amantadine (1-aminoadamantane hydrochloride) and Rimantadine (α -methyl derivative of amantidine)
 - Both are tricyclic amines of the adamantane family
- ❖ They are active against **influenza A only**.

MOA....block the **M2 proton** ion channel and inhibit **uncoating** of the viral RNA
...through this prevent replication.

Amantadine & Rimantadine

❖ Comparative activity

–Rimantadine = 4-10x active than amantadine *in vitro*.

PK of Amantadine

–Amantadine is well absorbed and

–Protein binding = 67% protein-bound

–plasma half-life = 12–18 hours

–Metabolism = It is excreted unchanged in the urine

PK of Rimantadine

–Protein binding = 40%

–half-life = 24–36 hours.

–Metabolism = metabolism by hydroxylation, conjugation, and glucuronidation before **urinary excretion**

❖Dose reductions are required for both agents in the elderly and in patients with **renal insufficiency**, and for **rimantadine** in patients with severe hepatic insufficiency

Amantadine & Rimantadine

❖ Both amantadine and rimantadine

- Protect **70–90%** in the prevention of clinical illness
- reduce the duration by **1–2** days when administered as treatment.

❖ due to high rates of resistance in both **H1N1** and **H3N2** viruses, these agents are **no longer** recommended for the **prevention** or **treatment** of **influenza**.

Adverse effects

❖ The most common adverse effects are

- gastrointestinal (nausea, anorexia) and
- central nervous system (nervousness, difficulty in concentrating, insomnia, lightheadedness)

- ❖ More serious side effects (eg, marked behavioral changes, delirium, hallucinations, agitation, and seizures) may be due to alteration of dopamine neurotransmission
 - less frequent with **rimantadine** than with **amantadine**
 - are associated with high plasma concentrations;
 - occur more frequently in patients with
 - renal insufficiency,
 - seizure disorders, or
 - advanced age; and
 - The above S/E (CNS) may increase with concomitant **antihistamines**, anticholinergic drugs, hydrochlorothiazide, and trimethoprim-sulfamethoxazole.

Other Antiviral Agents

Other antiviral agents: **Interferons**, Ribavirin, Palivizumab, Imiquimod

Interferons

In addition to **HBV** and **HCV** infections, interferons have been studied for numerous clinical indications

- intralesional injection of interferon **alfa-2b** or .
- may be used for treatment of condylomata acuminata

Ribavirin

Ribavirin is indicated for the following

- ✓ Given orally for HCV in combination with **interferon alfa**
aerosolized ribavirin
- ✓ In children and infants with severe respiratory syncytial virus (RSV) **bronchiolitis** or **pneumonia**
 - 20 mg/mL for 12–18 hours continuously per day) to reduce the severity and duration of illness.

Aerosolized ribavirin....

- ❖ Systemic absorption is low (<1%).
- ❖ Aerosolized ribavirin may cause
 - conjunctival or bronchial irritation and
 - may precipitate on contact lenses.

Palivizumab

- ❖ It is a humanized monoclonal antibody binds to a RSV surface glycoprotein **inhibit the F surface protein of RSV.**
- ❖ It is licensed for the prevention of RSV infection in
 - high-risk infants and children, such as premature infants and those with
 - bronchopulmonary dysplasia or congenital heart disease.

Other Antiviral Agents

❖ Imiquimod

❖ Imiquimod is an **immune response**

❖ Topical preparation (5% cream) is effective for the treatment of

- external genital and

- perianal warts (ie, condyloma acuminatum)

❖ Dosage:

❖ applied TID/weekl and washed off 6–10 hours after each application.

- Recurrences appear to be less common than after ablative therapies.

Imiquimod....

- ❖ Imiquimod may also be effective against **molluscum contagiosum (not licensed in the USA)**

Adverse effects

- ❖ Local skin reactions (the most common)
 - but tend to resolve within weeks after therapy.
 - However, pigmentary skin changes may persist.
 - Systemic adverse effects such as fatigue and influenza-like syndrome